KCWI/KCRM Counter-Rotating Disk Observing and Analysis Plan

Working technical summary of the observing strategy, injection-recovery tests, two-component pPXF decomposition, and global disk modeling

Prepared 8 August 2026

Status: blue-arm injection/recovery and fixed-light-fraction tests complete; RM2/RH3 red-arm injection/recovery experiment currently running.

Core conclusion so far: the KCWI Small+BL simulations recover the two-disk velocity separation very well at S/N around 40 per resolution element. The poorer absolute velocities in the original free-light-fraction fits are largely a light-fraction / velocity-midpoint degeneracy, not a lack of spectral resolution. When the true 50/50 light ratio is fixed, absolute velocity recovery rises to 95-96% at S/N 40-50 and median individual velocity errors fall to roughly 3 km/s. This motivates an explicit 3-D search in (V_A, V_B, f_A) for every Voronoi bin in the real cube rather than fixing the light fraction.

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Executive summary

The proposed KCWI/KCRM program is designed to turn a MaNGA counter-rotating-disk candidate into a spatially resolved two-component stellar-kinematic and stellar-population experiment. The central idea is to use the Small slicer so that the blue and red spectrograph arms observe the same 8.4 x 20.4 arcsec field simultaneously, with the blue arm providing broad stellar-population leverage and the red arm potentially providing exceptionally clean Ca II triplet kinematics.

The working primary target for the simulations is IC 25 (z = 0.01944). The injected residual velocities are V_A = -60 km/s and V_B = +69 km/s, giving a true separation of 129 km/s, with intrinsic dispersions sigma_A = sigma_B = 50 km/s and an equal-light baseline. The galaxy is approximately 25 arcsec across along the relevant kinematic axis; a single Small-slicer pointing therefore covers the central approximately 20 arcsec along one dimension and can be oriented so that the most informative counter-rotating region is inside the field. A mosaic remains optional if the science requires the outermost light distribution.

The current blue-arm recovery tests use the 4800-5500 A rest-frame region. With a free light ratio, the data recover Delta V very accurately but often shift both components together by approximately 15-20 km/s. A controlled experiment fixing f_A = f_B = 0.5 collapses that common-mode error to approximately 3 km/s and raises absolute-velocity recovery to 95-96% at S/N 40-50. This demonstrates that the original limitation is primarily covariance between light fraction and velocity midpoint. It also directly motivates the production analysis: explicitly grid V_A, V_B, and f_A in every Voronoi bin and save the full 3-D Delta-chi-square volume rather than relying on a single pPXF minimum.

For the final two-dimensional interpretation, the raw velocity maps for Disk A and Disk B can first be fit independently with XookSuut as a published circular-disk validation and to obtain starting values for the disk geometry. A custom joint optimizer can then use the same circular-disk prescription to regenerate both velocity fields while evaluating them against the original per-bin pPXF likelihood cubes. The global model is therefore constrained by the spectra themselves rather than by smoothing the raw maps.

1. Scientific objective and target context

The science goal is not merely to confirm counter-rotation. It is to recover two spatially coherent stellar disks, measure their independent velocity and velocity-dispersion fields, separate their light contributions, and then determine whether their stellar populations and spatial structures favor a particular formation pathway.

- Primary simulated target: IC 25, z = 0.01944. Working injected kinematics: (-60, +69) km/s; Delta V = 129 km/s; sigma_A = sigma_B = 50 km/s; baseline f_A = 0.50.
- Second target discussed in the planning: J164055, with representative injected residual velocities approximately (+60, -65) km/s. The refined simulations have intentionally focused on IC 25 first.
- MaNGA remains valuable for large-scale context: stellar and gas velocity geometry, H-alpha/[N II]/[S II]/[O III] emission-line information, BPT context, and the location of the 2-sigma morphology. KCWI/KCRM supplies the high spatial and spectral leverage needed for the decomposition itself.
- The formation-history information will come from the combination of kinematics, component light distributions, ages/metallicities, kinematic temperature (sigma/V), gas alignment, and radial structure. No single hard threshold should be used as an automatic “formation classifier.”

2. KCWI/KCRM observing plan

2.1 Small slicer and field placement

The Small slicer is favored because it gives the highest spectral resolution for every grating and 0.35 arcsec slice sampling. Keck lists a Small-slicer field of approximately 8.4 x 20.4 arcsec. The red KCRM arm is used simultaneously with the blue arm, shares the same field of view, and uses the same slicer. This is ideal for comparing the blue and red two-component decompositions because the two wavelength regimes sample the same physical area.

A practical pointing strategy is to orient the approximately 20 arcsec dimension along the stellar kinematic major axis so that the central two-sigma / maximum-separation region is well covered. For IC 25, this should contain the scientifically important approximately +/-8 arcsec region even though a single pointing does not cover the full approximately 25 arcsec galaxy diameter.

2.2 Blue arm: Small + BL

Quantity	**Working choice**	**Rationale / note**
Slicer	Small	0.35 arcsec slices; ~8.4 x 20.4 arcsec field
Grating	BL	Best broad blue coverage and efficiency
Nominal resolving power	R ~ 3600	sigma_inst ~ 35 km/s by c/(2.355R)
Instantaneous coverage	~2000 A	A central wavelength near 4600 A gives roughly 3600-5600 A
Simulation kinematic window	4800-5500 A rest	H-beta / Fe / Mg region; avoids requiring very high S/N at 4000-4500 A
Simulation sampling	~0.625 A/pixel; FWHM ~1.25 A	Based on the ETC-style setup used in the injection tests; final analysis must use the measured on-sky LSF

$$\sigma_{\text{inst}} \simeq \frac{c}{2.355\,R}$$

The blue arm is the broad-band stellar-population workhorse. The primary kinematic decomposition can be anchored in 4800-5500 A, while the bluer Balmer/4000-A diagnostics may be analyzed at lower S/N after the kinematics have been stabilized.

2.3 Red arm: RM2 versus RH3

The red-arm injection/recovery wrapper is currently running, so the document deliberately does not claim a final RM2-versus-RH3 S/N threshold. Instrumentally, however, both are substantially sharper than BL and both include the Ca II triplet in an appropriate setting.

Setup	R	Approx. sigma_inst	Instantaneous range	Primary role
BL (Small)	~3600	~35.4 km/s	~2000 A	Broad blue populations + kinematics
RM2 (Small)	>5600	<22.7 km/s	~1750-2000 A	Broad red kinematic region: Na I, CaT, Mg I, Fe/Ti
RH3 (Small)	>13000	<9.8 km/s	~750-850 A	Highest-resolution CaT kinematics

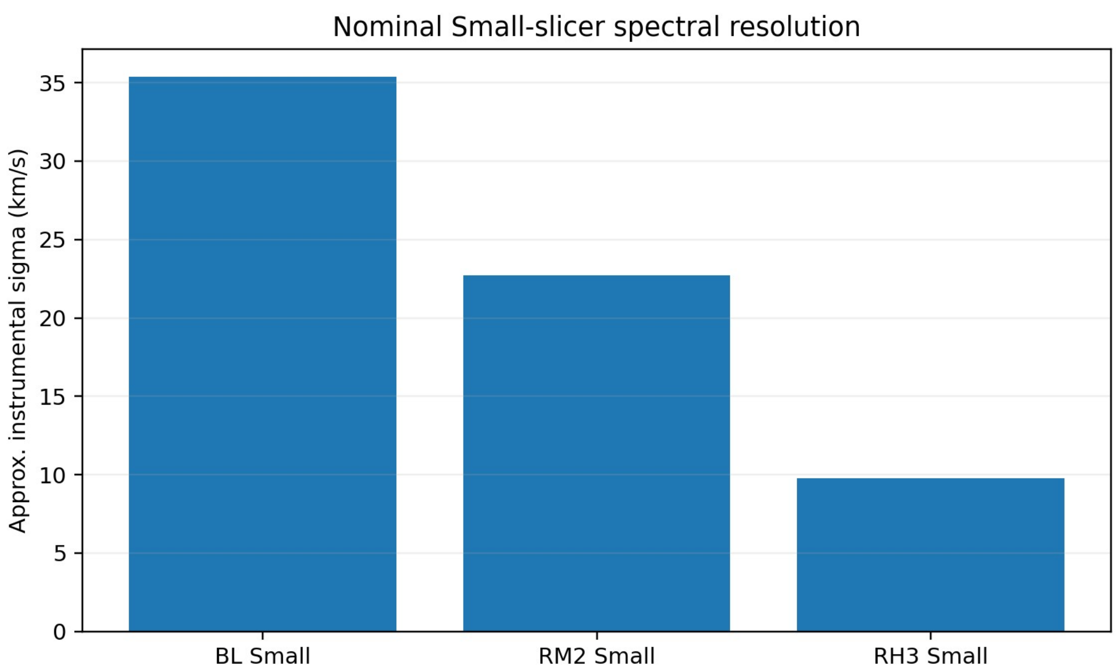

Figure 1. Nominal instrumental velocity dispersion for the Small slicer. RH3 makes instrumental broadening nearly negligible compared with an intrinsic stellar dispersion of ~50 km/s.

At sigma_star = 50 km/s, the approximate observed Gaussian width becomes sqrt(sigma_star^2 + sigma_inst^2): about 61 km/s for BL, 55 km/s for RM2, and 51 km/s for RH3. This is why RH3 could produce particularly clean V and sigma maps, although RM2 may compensate for its lower R by providing far more absorption-line coverage. The running injection/recovery experiment is intended to decide this empirically.

The red wrapper currently tests RM2 over a broad 7900-8950 A rest-frame window and RH3 over 8470-8900 A rest frame. The exact RM2 central wavelength must be confirmed with the configuration tool before observing. For RH3, Keck gives an example central wavelength of 9040 A with approximately 8630-9380 A observed coverage, comfortably containing the redshifted CaT for IC 25.

2.4 Simultaneous exposure cadence and stacking

All individual blue exposures will be registered and stacked into one final blue cube; the red exposures will likewise be stacked into one final red cube. The key point is that the red data do not require separate telescope time: they are accumulated simultaneously while the blue detector is integrating.

- For the Small slicer, keep the blue detector in the appropriate 1x1 state throughout the sequence; Keck warns that changing blue binning introduces elevated read noise for an extended period.
- The red detector is unusually thick and has abundant cosmic rays. Keck recommends <=600 s red exposures in 1x1 and at least three exposures per position. Repeated 300-600 s red integrations can therefore run during each longer blue integration.
- Because the Small field is largely occupied by the galaxy and moonlight/sky residuals are a concern, dedicated blank-sky observations should be treated as part of the observing plan rather than assuming the object cube contains sufficient uncontaminated sky.
- The full-Moon blue spectrum also contains scattered solar-spectrum structure. Sky subtraction quality, not only photon statistics, is therefore an important systematic in the blue.

2.5 Moon illumination and the blue exposure-time problem

Moon illumination by itself does not determine the exposure time. The lunar altitude, target altitude, angular separation from the target, atmospheric extinction, and wavelength matter strongly. The table below is therefore a planning-level comparison, not an ETC prediction. It uses the Krisciunas & Schaefer (1991) Mauna Kea moonlight model as a V-band proxy at the representative IC 25 geometry discussed earlier: target altitude ~54 deg, Moon altitude ~63 deg, and Moon-target separation ~29 deg.

The absolute exposure-time scale is tied to the earlier working estimate of 6.4 h to reach S/N = 50 at 4500 A for a sky surface brightness near 19.5 mag/arcsec^2. Pure S/N scaling gives 4.10 h for S/N = 40 at the same background. In the background-dominated limit, the additional sky-brightness scaling is approximately:

$$t_2 \approx t_1 \left(\frac{S/N_2}{S/N_1}\right)^2 10^{0.4\,(\mu_{\mathrm{sky},1} - \mu_{\mathrm{sky},2})}$$

Moon illumination	Phase angle	Approx. sky mu_V	Time factor vs mu=19.5	Scaled t for S/N=40 at 4500 A
0%	180.0 deg	21.71	0.13x	0.54 h
25%	120.0 deg	20.91	0.27x	1.12 h
50%	90.0 deg	20.01	0.62x	2.56 h
75%	60.0 deg	19.16	1.36x	5.58 h
99%	11.5 deg	17.93	4.26x	17.46 h

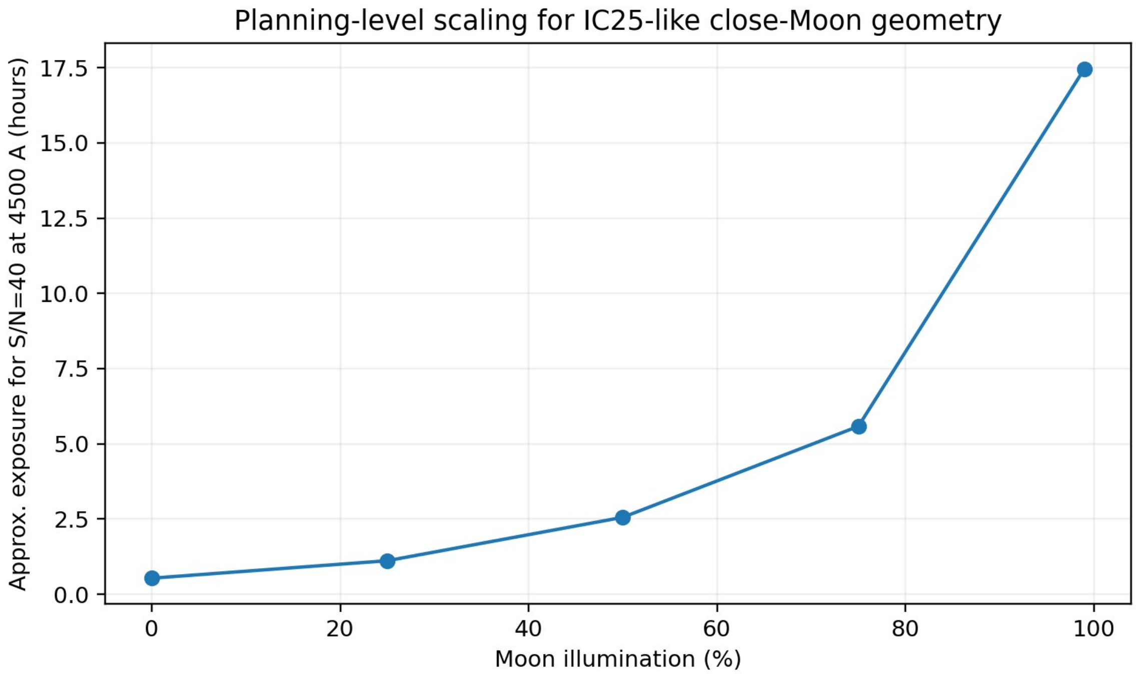

Figure 2. Illustrative blue exposure-time scaling for the IC 25-like close-Moon geometry. The steep rise at high illumination is the main practical point; exact values require the current KCWI ETC and the actual lunar geometry.

At 99% illumination and only ~29 deg separation, the illustrative requirement is ~17.5 h to reproduce the same S/N = 40 at 4500 A. At the ~16 deg separation discussed for the other science target, the same calculation is even worse (~25.7 h). These values should not be read as final requested times; they show why requiring S/N = 40 at 4500 A under a nearly full nearby Moon is unrealistic.

The practical bright-time strategy is therefore to define the kinematic S/N requirement in the redder 4800-5500 A BL window, where throughput is better and lunar scattering is less severe than at 4000-4500 A, and to accept lower S/N in the bluest population diagnostics. Once the two LOSVDs are fixed using the strongest kinematic information, the lower-S/N blue features can still contribute population information when co-added in the same Voronoi bins.

3. Blue-arm injection-recovery results and the S/N requirement

3.1 Refined baseline experiment

The refined simulation is deliberately conservative in one important sense: the two injected disks have identical stellar populations, so the only information distinguishing them is their LOSVD. The baseline IC 25 injection uses age ~8 Gyr, [M/H] ~-0.2, V_A = -60 km/s, V_B = +69 km/s, sigma_A = sigma_B = 50 km/s, and f_A = 0.50.

- S/N grid: 30, 35, 40, 45, 50 per spectral resolution element.
- 100 two-component Monte Carlo realizations and 50 one-component null realizations at each S/N and template case.
- 17 x 17 velocity grid from -160 to +160 km/s in 20 km/s steps, giving 289 pPXF cells per spectrum. Within each cell, each velocity is bounded to +/-10 km/s around the grid center while sigma remains free from 5-180 km/s.

- Kinematic continuum freedom: additive polynomial degree 4, multiplicative degree 0 in the simulation.
- One-component null: V = 0 km/s, sigma = 90 km/s. The 95th percentile of null Delta chi^2 defines the empirical two-component significance threshold.

3.2 Free-light-fraction result

With the light fractions free, the spectral decomposition clearly sees two LOSVDs and recovers their separation, but it often allows both recovered velocities to translate together. At S/N = 40 in the matched-control case, the detection fraction is 0.84 and relative-kinematic recovery is 0.82, but strict absolute-velocity recovery is only 0.46. The median separation error is only ~3.6 km/s while the median common midpoint offset is ~18.6 km/s.

S/N	Detection	Relative DeltaV	Absolute velocities	Median |dDeltaV|	Median common offset
30	0.53	0.50	0.29	6.4	19.4
35	0.55	0.54	0.34	5.3	19.2
40	0.84	0.82	0.46	3.6	18.6
45	0.86	0.85	0.54	3.0	17.0
50	0.85	0.85	0.51	3.0	17.2

The matched-control and sparse mismatched-basis results are nearly identical. At S/N = 40 the detection fractions are 0.84 versus 0.83; at S/N = 50 they are both 0.85. Because the matched control contains the exact injected SSP, this argues strongly that the limited eight-template basis is not the primary source of the common-mode velocity floor.

3.3 Fixed-light-fraction diagnostic

The decisive diagnostic fixed the known injected fraction to f_A = f_B = 0.50 while leaving V_A, V_B, sigma_A, sigma_B, and the SSP mixture within each component free. This removes only the fraction-midpoint degeneracy.

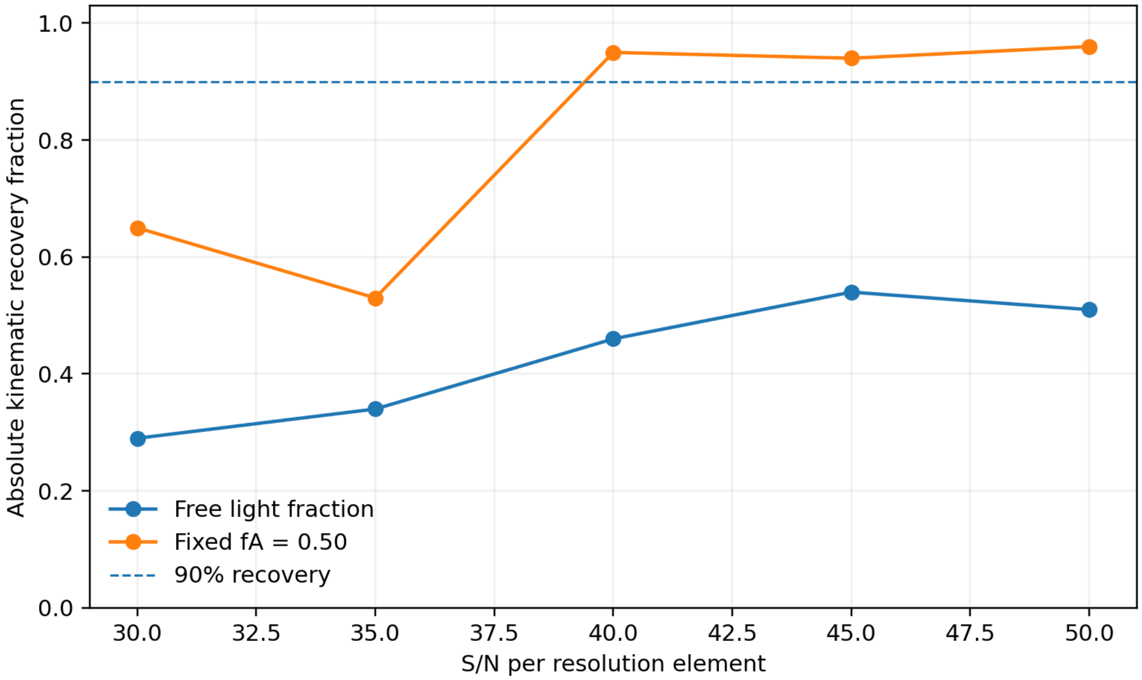

Figure 3. Absolute velocity recovery changes dramatically when the correct 50/50 light ratio is fixed. The experiment isolates the light-fraction / velocity-midpoint degeneracy.

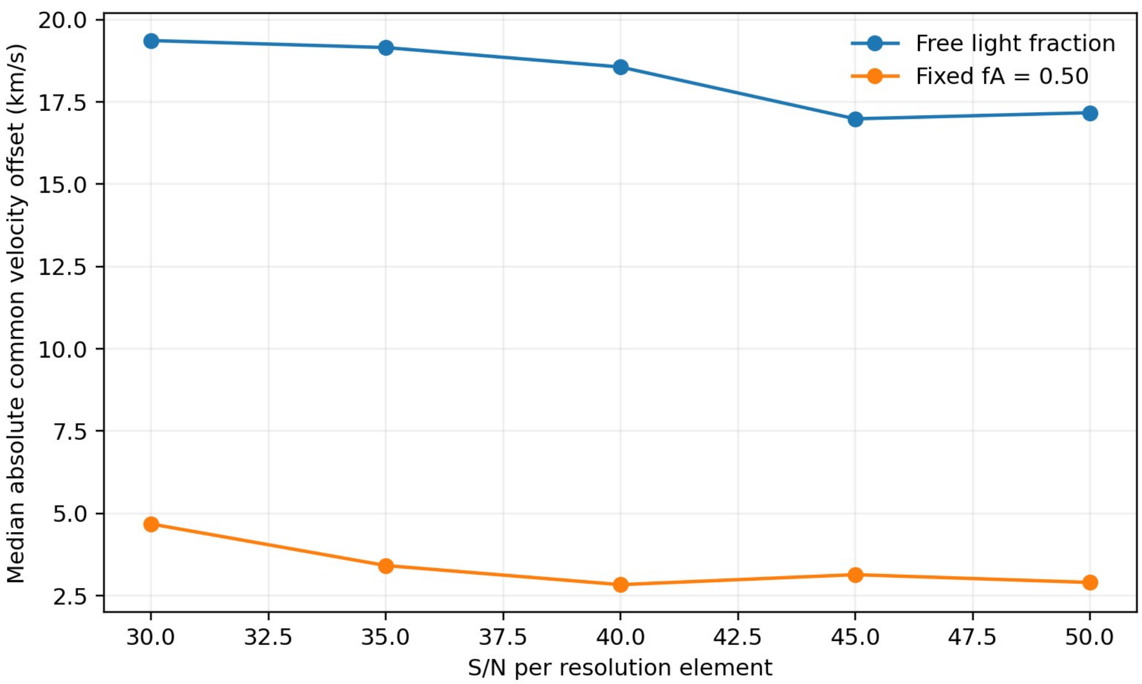

Figure 4. The median common velocity offset collapses from roughly 17-20 km/s to roughly 3-5 km/s when the light fraction is fixed.

S/N	Absolute recovery	Both V within 20	Median |dV_A|	Median |dV_B|	Median common offset
30	0.65	0.98	4.5	4.7	4.7
35	0.53	0.99	4.3	4.0	3.4

40	0.95	0.99	3.6	3.1	2.8
45	0.94	1.00	3.5	3.6	3.1
50	0.96	1.00	2.8	3.3	2.9

At S/N = 40 the strict absolute-recovery fraction rises from 0.46 to 0.95; the median individual velocity errors fall from about 19 and 17 km/s to about 3.6 and 3.1 km/s; and the common offset falls from ~18.6 to ~2.8 km/s. At S/N = 45-50 the fixed-fraction absolute recovery is 94-96%. This is strong evidence that the data contain enough spectral information for accurate individual velocities and that the earlier “poor” absolute fits are primarily caused by covariance with f_A.

The non-monotonic S/N = 35 point is driven by the empirical null significance threshold rather than by a sudden loss of velocity information: the absolute-velocity pass fraction there is 0.99, while the significance pass fraction is only 0.53. For publication-quality significance calibration, a larger null ensemble (for example 200-1000 realizations or an uncertainty analysis of the tail quantile) is advisable.

3.4 What this implies for the observing S/N target

The current working target is S/N ~40 per resolution element in the primary BL kinematic window. In the simulation this corresponds to ~28.3 per spectral pixel because the adopted BL model has approximately two pixels per resolution element. S/N = 40 is not a magic threshold for every spatial bin, but it is the point where the baseline detection/relative recovery rises sharply and the fixed-fraction velocity information becomes extremely strong.

The correct conclusion is not to fix the real data to 50/50. Instead, the fixed-fraction experiment tells us to measure the fraction explicitly. The production fit should therefore scan f_A as a third grid dimension so that the full covariance among V_A, V_B, and f_A is visible rather than being hidden inside the linear template weights.

3.5 Red-arm simulation status

The RM2/RH3 wrapper currently runs both free-fraction and fixed-50/50 versions, uses the matched-control basis, extends the S/N grid down to 10-50 per resolution element, and adds explicit sigma_A/sigma_B recovery diagnostics. Its first pass uses 20 null and 50 CRD realizations per S/N; a high-statistics switch restores 50 null and 100 CRD realizations after the transition region is identified. The results are pending and should determine whether RM2 or RH3 is the better primary kinematic channel.

4. Proposed two-component pPXF decomposition

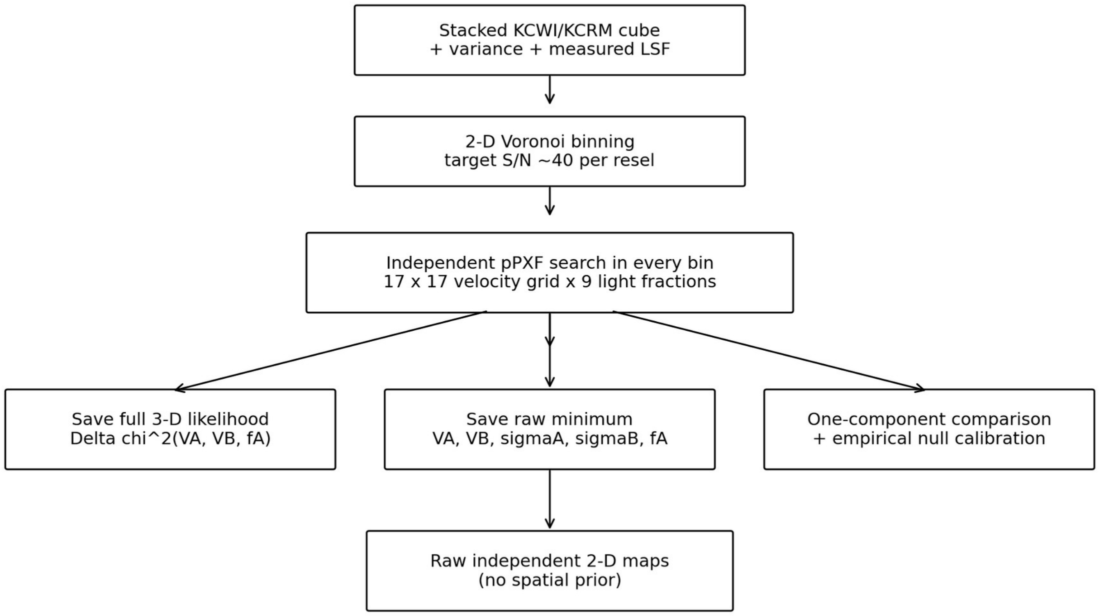

Figure 5. Proposed production pPXF workflow for each stacked KCWI/KCRM cube.

4.1 Cube preparation and Voronoi binning

1. Reduce and stack all individual exposures into one blue cube and one red cube, preserving variance and covariance information as well as a measured wavelength-dependent LSF.
2. Register the two arms spatially and define a common astrometric reference. For direct blue/red comparisons, match the PSF or forward-model the different PSFs.
3. Construct a continuum S/N map and Voronoi-bin the cube in two dimensions. The current expectation is roughly 100-200 useful bins per cube. The target S/N for the primary blue kinematic region is approximately 40 per resolution element.
4. Keep the bin membership map so every recovered quantity can be displayed over the correct native spaxels while retaining one statistically independent spectrum per Voronoi bin.

4.2 Templates, LSF, continuum, and emission lines

Every stellar template must be matched to the actual instrumental LSF before interpreting the pPXF sigma as the intrinsic stellar dispersion. The production analysis should use the measured wavelength-dependent KCWI/KCRM LSF rather than a nominal constant FWHM.

The real analysis should fit nebular emission templates where practical instead of simply masking H-beta or other important stellar-population lines. Continuum polynomials should remain modest. The simulations use additive degree 4 and no multiplicative polynomial; degree 0/2/4 and modest multiplicative alternatives should be tested for systematic shifts in the velocity midpoint.

A full XSL/SSP library is important for the final stellar-population analysis, but the matched-versus-mismatched simulation shows that adding the exact injected SSP does not cure the free-fraction common-mode velocity floor. More templates are therefore not a substitute for explicitly handling f_A.

4.3 The explicit 3-D brute-force grid

The production search should calculate a genuine three-dimensional likelihood volume in each Voronoi bin: V_A x V_B x f_A. A practical first grid is 17 x 17 velocities and nine fractions f_A = 0.1, 0.2, ..., 0.9. This is 2601 pPXF fits per bin. At 100-200 bins this is ~260,000-520,000 fits per cube; for two cubes the total remains an overnight-scale calculation based on the measured throughput of the current scripts.

$$F_{\mathrm{model}}(\lambda) = \sum_j w_{A,j} T_j(V_A, \sigma_A) + \sum_j w_{B,j} T_j(V_B, \sigma_B)$$

$$f_A + f_B = 1, \qquad f_B = 1 - f_A$$

There is only one independent fraction parameter: once f_A is specified, f_B = 1 - f_A. pPXF still fits two independent velocities because no constraint requires V_B = -V_A. At every grid point, the light ratio is fixed while sigma_A, sigma_B and the template mixture remain free. A later refinement can use f_A spacing of 0.05 or a coarse-to-fine grid around the best fraction cells.

For each bin, save the entire Delta chi^2(V_A,V_B,f_A) array, not just the global minimum. The complete cube is inexpensive to store and is essential for recognizing broad valleys, alternate minima, and the covariance that the fixed-fraction experiment exposed.

4.4 Raw minimum maps and label symmetry

Before imposing any spatial model, take the independent global minimum of every bin and construct raw maps of V_A, V_B, sigma_A, sigma_B, f_A, Delta V, and fit significance. These maps answer the cleanest possible question: what does each spectrum prefer with no spatial information?

The component labels are intrinsically symmetric. The solution (V_A, V_B, f_A) has an equivalent relabeling (V_B, V_A, 1-f_A). The first pass should therefore retain neutral component-1/component-2 labels, and only afterward assign physical Disk A/B labels using global coherence and a documented convention (for example, Disk A is the component co-rotating with the ionized gas). Every dependent quantity must be swapped together.

4.5 One-component comparison and real-data detection

The injection tests can define success using the known truth, but the real data cannot. Real-data detection should instead combine: (1) improvement over the best one-component fit, calibrated empirically; (2) a meaningful light contribution from both components; (3) a resolved/non-zero separation and a localized likelihood structure rather than a flat ridge; and (4) spatial coherence as a secondary validation rather than a primary detection assumption.

A stronger null suite should eventually vary the one-component intrinsic sigma (for example 50, 70, 90, 110 km/s), include realistic sky/telluric residuals, and use a larger null Monte Carlo sample. This prevents the significance criterion from being tuned to only one broad single-LOSVD case.

5. Spectral diagnostics in the blue and red arms

5.1 Blue BL diagnostics

Feature	Rest wavelength (A)	Primary use

Ca H&K	3934, 3968	Age/metallicity leverage; blue S/N and sky systematics important
D_n4000	~3850-4100 bands	Age-sensitive continuum break
H-delta	4102	Age / recent-star-formation sensitivity
G band	~4304	Stellar population / metallicity leverage
H-gamma	4340	Age-sensitive Balmer absorption
H-beta	4861	Age-sensitive; fit gas emission simultaneously
Fe5015	~5015	Metallicity / Fe-sensitive
Mg b	5167, 5173, 5184	Metallicity / Mg-sensitive; strong kinematic structure
Fe5270, Fe5335	5270, 5335	Fe-sensitive metallicity diagnostics

The 4800-5500 A interval is especially useful for the initial kinematics because it contains H-beta/Fe5015/Mg b/Fe features and is less moon-sensitive than the extreme blue. The 3900-4600 A region is retained for population information even if its per-bin S/N is lower.

5.2 Red RM2/RH3 diagnostics

Feature	**Rest wavelength (A)**	**Role / caution**
Na I doublet	~8183, 8195	RM2 only depending on exact setting; population/gravity sensitivity
Ca II triplet	8498, 8542, 8662	Strong red stellar absorption; primary red kinematic anchor
Mg I	8807	Additional red metallic-line / kinematic leverage
Fe/Ti forest	various	Additional line information, especially for RM2 broad coverage
Paschen lines	near CaT region	Potential contamination / population information; model or mask appropriately
OH + tellurics	red sky spectrum	Major red-arm systematic; requires careful sky/telluric handling

For IC 25 at z = 0.01944, the CaT lines appear at approximately 8663, 8708, and 8831 A observed; Mg I 8807 appears near 8978 A. These positions fit comfortably within the RH3 example setting centered at 9040 A. The CaT should be treated primarily as a kinematic and additional metallicity diagnostic, not as a single-parameter abundance meter because it is also sensitive to surface gravity, abundance pattern, and the IMF.

6. From pPXF likelihood cubes to two coherent disks with XookSuut

The global modeling has two distinct stages. Stock XookSuut is run only twice - once on the raw Disk-A velocity map and once on the raw Disk-B velocity map. Those fits are useful independent validations and provide excellent starting values for the geometry. The later joint fit is our own optimizer using the same published circular-disk prescription; it is not a third XookSuut run.

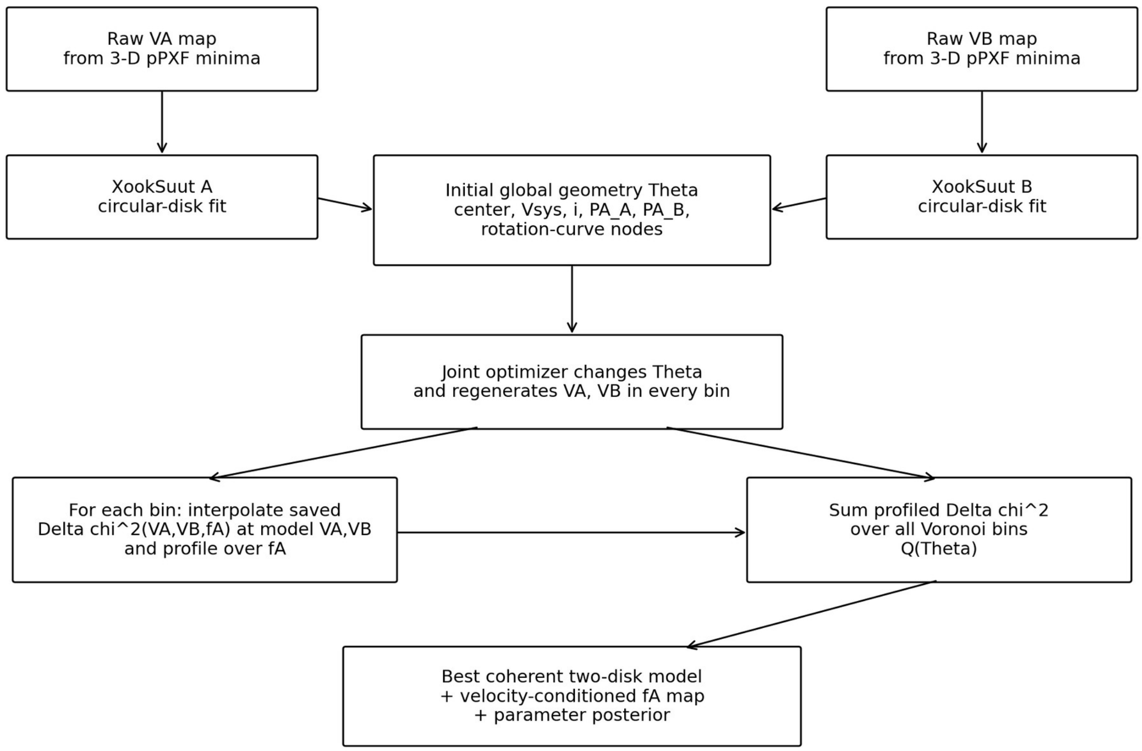

Figure 6. Relationship between the raw pPXF maps, the two independent XookSuut fits, and the custom joint likelihood optimization.

6.1 Circular-disk prescription

For a thin disk in circular rotation, the line-of-sight velocity at sky position (x,y) is:

$$V_{\rm los}(x,y) = V_{\rm sys} + V_{\rm rot}(R)\sin i\cos\theta$$

Given a trial center (x_0,y_0) and position angle PA, translate and rotate the sky coordinates. Schematically, with the exact sign convention taken from XookSuut when implemented:

x' = Delta x cos(PA) + Delta y sin(PA);   y' = -Delta x sin(PA) + Delta y cos(PA)

Deproject the minor-axis coordinate by the inclination i to obtain the disk-plane radius and azimuth:

$$R = \sqrt{x\prime^2 + \left(\frac{y\prime}{\cos i}\right)^2}, \qquad \cos\theta = \frac{x\prime}{R}$$

The rotation speed V_rot(R) can be represented by radial nodes/rings and interpolated between them, matching the DiskFit/XookSuut philosophy. Changing PA, inclination, center, systemic velocity, or a rotation-curve node changes the predicted velocities in many or all Voronoi bins simultaneously.

6.2 Initial independent XookSuut fits

Run XookSuut separately on the raw V_A and V_B maps. These two runs test whether the completely independent pPXF minima already behave like rotating disks under a published 2-D velocity-field model. Compare the fitted centers, systemic velocities, PAs, inclinations, and rotation curves. Kinemetry and fit_kinematic_pa can provide independent checks of the axis and non-circular residuals.

For the first joint model, a reasonable parameterization is to share (x_0, y_0, V_sys) and probably inclination i, while initially allowing PA_A and PA_B and the two radial rotation curves to be independent. The data can then test whether the PAs are consistent and whether a more constrained coplanar model is justified.

6.3 The custom joint likelihood

Let Theta denote the global disk parameters. For every trial Theta, the circular-disk equations generate V_A(x_i,y_i|Theta) and V_B(x_i,y_i|Theta) at the position of every Voronoi bin. We then return to the original saved pPXF Delta-chi-square cube for that bin.

At the model-predicted velocity pair, interpolate the 3-D cube and profile over the fraction direction: choose the f_A that gives the lowest Delta chi^2 at those velocities. The objective function is:

$$Q(\Theta) = \sum_i \min_{f_A} \Delta\chi_i^2[V_A(x_i, y_i \mid \Theta),\ V_B(x_i, y_i \mid \Theta),\ f_A]$$

This is the quantity that is minimized or sampled. We are not minimizing the sum of f_A values. We are minimizing the sum of the spectral penalties after every bin has been allowed to choose the light fraction that best fits its own spectrum at the globally coherent pair of velocities.

A noisy outer bin with a broad likelihood surface contributes little discrimination because many model velocities have similar Delta chi^2. A high-S/N bin with a narrow minimum contributes strongly. This is preferable to simply fitting a smooth curve through the raw minima or manually selecting only high-S/N bins.

6.4 Final velocity-conditioned light-fraction map

Once the best global two-disk geometry has been found, the preferred light fraction in each bin is obtained by profiling the same cube at the final model velocities:

$$f_{A,i}^{\mathrm{final}} = \arg\min_{f_A} \Delta\chi_i^2\left(V_{A,i}^{\mathrm{final}}, V_{B,i}^{\mathrm{final}}, f_A\right)$$

This produces a velocity-conditioned f_A map without forcing f_A itself to be spatially smooth. It can be compared directly with the raw independent f_A map. A later structural model can then test whether the resulting component light maps are consistent with smooth disks.

6.5 Optional Stage 3: structural light model

If the conditioned light map is coherent, define component surface-brightness maps from the total continuum intensity:

$$I_A = f_A I_{\rm tot}, \qquad I_B = (1 - f_A) I_{\rm tot}, \qquad f_A^{\rm model} = \frac{I_A}{I_A + I_B}$$

The simplest physical model is two exponential disks I(R) = I_0 exp(-R/h), potentially sharing a center but with separate scale lengths, normalizations, ellipticities, and PAs. This turns the light fractions into structural measurements rather than an arbitrary smoothed scalar field. It can test whether one counter-rotating component is more centrally concentrated, thinner, or more extended.

6.6 Bayesian inference and validation

A simple scipy optimizer is useful for code validation, but the final analysis should return uncertainties and parameter covariances. XookSuut itself uses MCMC and nested sampling; the custom likelihood can similarly be explored with emcee or dynesty. The output should include posterior intervals for PA, inclination, V_sys, rotation-curve nodes, and any shared geometry parameters.

- Kinemetry: quantify kinematic PA versus radius, rotation amplitude, twists, and higher-order deviations independently of the circular model.
- fit_kinematic_pa: simple independent global PA measurement for each raw and modeled velocity field.
- Residual maps: show V_raw - V_model and flag coherent residual structure that may require warps, radial flows, or a more flexible model.
- PSF / Voronoi integration: the final forward model should account for flux-weighted PSF convolution and the actual footprint of each Voronoi bin, especially near steep central velocity gradients.

6.7 Combining the blue and red cubes

The blue and red arms should trace the same underlying stellar velocities but need not have the same light fractions because the two stellar populations can have different colors. Thus the physically sensible joint model shares V_A and V_B between arms while allowing f_A^blue and f_A^red to differ. Agreement between independently decomposed blue and CaT velocity fields would be a strong validation that the two recovered disks are physical rather than template artifacts.

7. Stellar populations and formation-history constraints

7.1 Separate kinematics from population inference

The safest sequence is to establish the two-component kinematics first and then hold or tightly constrain those LOSVDs while measuring population information. The simultaneous two-component pPXF regularization route is not ideal because pPXF regularization is not symmetric across arbitrary multiple components in the way needed here. Instead, reconstruct the two component model spectra and analyze their populations separately after the kinematics are established.

- Initial kinematics: unregularized, modest continuum polynomials, gas templates included where needed.
- Component spectra: reconstruct F_A(lambda) and F_B(lambda) from the fitted template contributions rather than treating raw template-weight sums as the final physical light fraction.
- Population fits: fit each reconstructed component spectrum with the full XSL/SSP population basis; use regularization only in a clearly defined one-component population fit if desired.

- Report light fractions in an explicit wavelength band because f_A^blue and f_A^red can differ even when the stellar mass fractions are the same.

7.2 Population diagnostics

Blue diagnostics should include H-beta, H-gamma, H-delta, D_n4000, Mg b, Fe5270, Fe5335, and potentially Fe5015/G4300/Ca H&K depending on S/N and calibration. A useful metallicity-sensitive composite is:

$$[\mathrm{MgFe}]' = \sqrt{\mathrm{Mg}b\,(0.72\,\mathrm{Fe5270} + 0.28\,\mathrm{Fe5335})}$$

The red arm adds CaT, Mg I 8807, Na I for RM2 where available, and weaker Fe/Ti structure. These should be interpreted with SSP models rather than converted into a hard abundance classification.

7.3 Formation-scenario observables

Observable	Question	Formation leverage
Age of A vs B	Is one disk younger? Is the difference radial?	A younger gas-corotating secondary can support external gas accretion followed by in-situ star formation
[M/H] and Mg/Fe-sensitive behavior	Are the disks chemically distinct?	Different enrichment histories help distinguish accretion timescale and progenitor gas properties
sigma/V and thickness proxy	Which disk is dynamically colder?	A young secondary formed from settled accreted gas may be colder/thinner
Scale length / light distribution	Which component is more extended or concentrated?	Structural differences constrain whether the secondary was built gradually or deposited violently
Gas alignment	Which stellar component co-rotates with ionized gas?	A gas-corotating younger component is a classic external-gas-accretion signature
Blue vs red light fraction	Does the component ratio change with wavelength?	Direct evidence for color/population differences between the disks
2-D twists/asymmetries	Are the disks coplanar and axisymmetric?	Strong non-circular structure may point toward a more complex interaction/merger history

The precedent is strong: Coccato et al. separated counter-rotating components and their line strengths in NGC 3593/4550; Pizzella et al. used MUSE including the CaT to map both disks in IC 719 and found distinct kinematics, ages, metallicities, and spatial distributions; Bao et al. compared pPXF, Ca II 8542 double-Gaussian decomposition, and orbit-based modeling and obtained consistent two-disk kinematics. The KCWI/KCRM program extends this logic by retaining a full per-bin 3-D spectral likelihood and tying it to a global two-disk model.

8. Validation, systematics, and simulations still needed

Test / issue	Status	Why it matters
Red-arm RM2/RH3 recovery	Currently running	Do not choose the final red grating solely from nominal R; compare V, DeltaV, and

		sigma recovery at matched S/N.
Full fraction grid in injections	Next	Repeat injection/recovery with explicit f_A grid to quantify the covariance surface directly.
Unequal light ratios	Next	Test f_A ~0.35 and 0.20; equal light is favorable and not representative of every radius.
Unequal sigmas	Next	Test combinations such as 40/70 km/s.
Different populations	Next	Use old+young injections; population differences may help break the light-ratio degeneracy.
Full XSL / alternative injection library	Next	Matched-control result says modest mismatch is not dominant, but realistic cross-library mismatch should still be tested.
Grid resolution	Targeted test	Use 10 km/s velocity steps or local refinement around the best cells to rule out grid-boundary effects.
Realistic moon/sky noise	Important	Inject wavelength-dependent sky, scattered solar spectrum, OH/telluric residuals, correlated cube noise, and LSF errors.
Null significance tail	Publication-quality upgrade	Increase null realizations and test multiple one-component sigmas.
PSF / beam smearing	Final forward model	Convolve flux-weighted component models before comparing to central Voronoi measurements.

9. End-to-end analysis workflow and expected paper products

5. Reduce, sky-subtract, register, and stack all blue exposures into one BL cube and all simultaneous red exposures into one KCRM cube; characterize LSF and PSF.
6. Voronoi-bin each cube to the chosen S/N target while preserving bin footprints and variance/covariance information.
7. Run a one-component pPXF fit for quality control, systemic velocity context, optimal-template construction, and comparison against the two-component model.
8. Run the independent 3-D pPXF grid in every bin: V_A x V_B x f_A, with sigma_A/sigma_B and template mixtures free. Save the complete Delta chi^2 cube and all best-fit products.
9. Construct raw independent maps of V_A, V_B, sigma_A, sigma_B, f_A, DeltaV, and fit significance. Resolve component-label symmetry globally but do not change the measured values.
10. Run XookSuut independently on the two raw velocity maps and use Kinemetry/fit_kinematic_pa as validation.
11. Initialize a custom joint two-disk optimizer from the XookSuut solutions. Regenerate V_A and V_B from the circular-disk prescription at every optimizer step; evaluate every per-bin pPXF cube and profile over f_A; minimize/sample the summed spectral penalty.
12. Produce the final coherent Disk-A/Disk-B velocity fields and the velocity-conditioned f_A map; compare raw and modeled maps and inspect residuals.

13. Fix/constrain the validated kinematics and recover component spectra and stellar-population information separately in the blue and red arms.
14. Combine ages, metallicities, line indices, sigma/V, component scale lengths, blue/red light fractions, gas alignment, MaNGA large-scale context, and asymmetry measurements into a formation-history argument.

Expected high-value paper figures:

- BL raw and globally modeled V_A and V_B maps; KCRM RM2/RH3 equivalent maps.
- Disk-A and Disk-B sigma maps, especially if RH3 delivers the expected high precision.
- Raw and velocity-conditioned f_A maps, plus component surface-brightness models.
- Representative 3-D likelihood slices showing a sharp minimum and a fraction-midpoint degeneracy valley.
- Blue-versus-red velocity comparison for matched Voronoi bins.
- Rotation curves for both disks with posterior uncertainties and kinematic PA/inclination constraints.
- Component age, metallicity, and diagnostic-index maps/profiles.
- MaNGA gas map overlaid or compared with the disk that co-rotates with the gas.

Appendix A. Numerical recovery results

The tables below collect the most important current blue-arm results. “Free f_A” uses the normal two-component fit in which pPXF can redistribute the component weights. “Fixed f_A” forces the known injected 50/50 light ratio and is a diagnostic of the velocity-midpoint degeneracy.

S/N	Abs. rec. free	Abs. rec. fixed	Common off. free	Common off. fixed	|dDeltaV| free	|dDeltaV| fixed
30	0.29	0.65	19.4	4.7	6.4	5.4
35	0.34	0.53	19.2	3.4	5.3	4.4
40	0.46	0.95	18.6	2.8	3.6	3.4
45	0.54	0.94	17.0	3.1	3.0	2.8
50	0.51	0.96	17.2	2.9	3.0	3.0

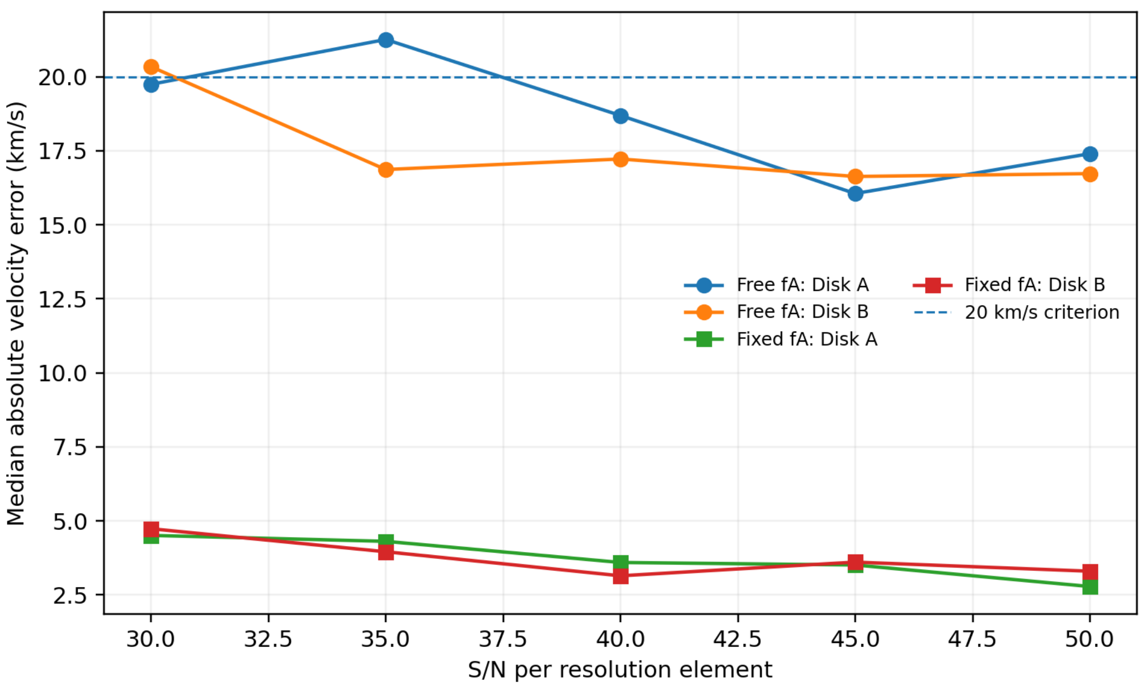

Figure A1. Median individual velocity errors for the free- and fixed-light-fraction experiments.

Null thresholds are themselves Monte Carlo estimates. In the original free-fraction matched-control run the 95th-percentile null Delta chi^2 thresholds were approximately 9.54, 10.00, 7.71, 8.40, and 10.67 for S/N 30-50. In the fixed-fraction run they were approximately 7.98, 9.17, 5.92, 7.08, and 8.52. Their non-monotonicity is another reason to enlarge the null sample for final significance calibration.

Appendix B. Moonlight exposure-time scaling assumptions

The illumination table is intended to show order-of-magnitude sensitivity, not to replace the ETC. The Krisciunas & Schaefer moonlight model was developed using Mauna Kea data and predicts sky brightness as a function of lunar phase, Moon zenith distance, target zenith distance, Moon-target angular separation, and extinction. For the illustrative IC 25 calculation we adopted target altitude ~54 deg, Moon altitude ~63 deg, separation ~29 deg, V-band extinction ~0.17 mag/airmass, and a dark zenith sky near V = 21.9 mag/arcsec^2.

The previous working exposure anchor was 6.4 h for S/N = 50 at 4500 A at mu_sky ~19.5 mag/arcsec^2. Scaling to S/N = 40 gives 4.096 h at that same background. In the sky-dominated limit, exposure time is proportional to sky flux. The resulting numbers are therefore especially uncertain at the lowest sky backgrounds, where source photon noise/read noise and the exact target spectrum become less negligible.

Because 4500 A is bluer than V, scattered moonlight can be at least as problematic as the V-band proxy suggests. Conversely, the actual 4800-5500 A kinematic window should be less severely affected than a 4500-A benchmark. A final observing request should be based on the current KCWI ETC plus the actual target surface-brightness spectrum and the real Moon geometry for the scheduled night.

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